

Best known for patenting the first telephone, British-born engineer Alexander Graham Bell also invented the photophone (a wireless communication device), refined the phonograph (which became the record player), and conducted groundbreaking work in the fields of surgery and nautical engineering.

MILESTONES

AWARDED PATENT

Receives a patent in 1876 for his technology that enables speech to be transmitted over wires.

PUBLIC APPEARANCE

Demonstrates his telephone publicly in 1876, making a call to a location 10 miles (16 km) away.

STARTS COMPANY

Founds the Bell Telephone Company in 1877, with financial backing from his father-in-law.

WIRELESS PHONE

Invents the photophone in 1880. The device transmits speech wirelessly using light.

In 1874, Alexander Graham Bell, a teacher to the deaf, began work on a device that, like the telegraph, used wires to transmit speech over long distances. Several other inventors were working toward the same goal—including Elisha Gray, who developed a liquid transmitter that Bell then adapted for his own experiments—but Bell beat them to producing and patenting a practical telephone. After numerous experiments with Thomas Watson, whom he recruited to help engineer his ideas, Bell discovered that the combination of a single steel reed relay vibrating over rotating magnets generated a current powerful enough to reproduce sound over a distance. It was this innovation that made telegraphic speech possible.

After further experimentation, Bell devised an electromagnetic transmitter, which he incorporated into his Butterstamp phone (named for its resemblance to the tool used for making butter pats). Patented in 1876, this was the first commercially viable telephone. But it was not the last of Bell's inventions. In 1877, he improved Thomas Edison's phonograph, which both recorded and replayed sound. He then developed a metal detector that could locate bullets in the human body—a device that was widely used in the Boer War and World War I. He also created the first hydrofoil; co-founded the National Geographic Society; and designed the tetrahedral space frame, forerunner of a construction technique that would revolutionize architecture and engineering in the 20th century.

Alexander Graham Bell's experimental telephone used a single electromagnet to turn electricity into sound. It could function as both a transmitter and a receiver.

“Mr. Watson, come here. I want to see you.”

Alexander Graham Bell, speaking the first words by telephone on March 10, 1876

